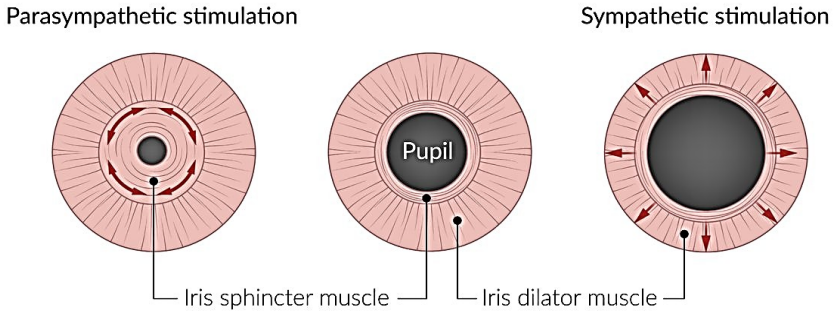


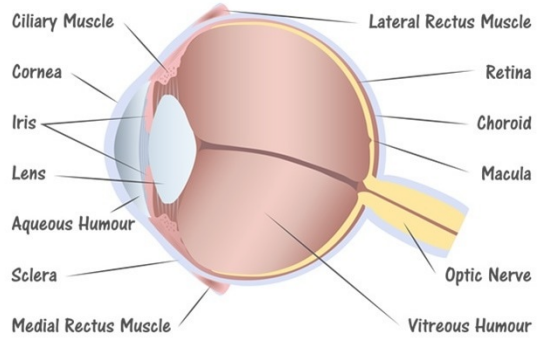
EFFECT OF DRUGS ON THE IRIS

PHARMACOLOGY LAB 5

Nermin Jamal Shaymaa Faruq

Academic year 2026–2027

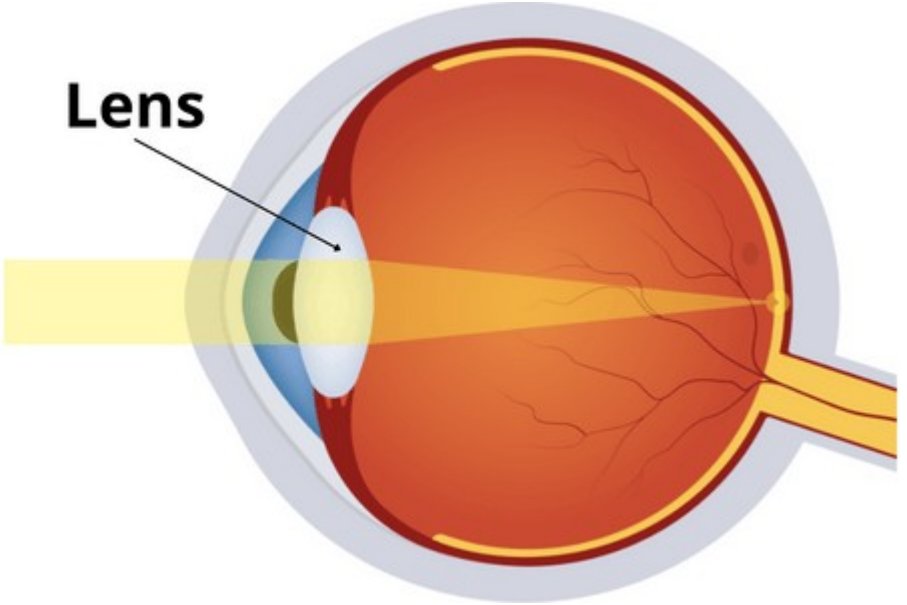




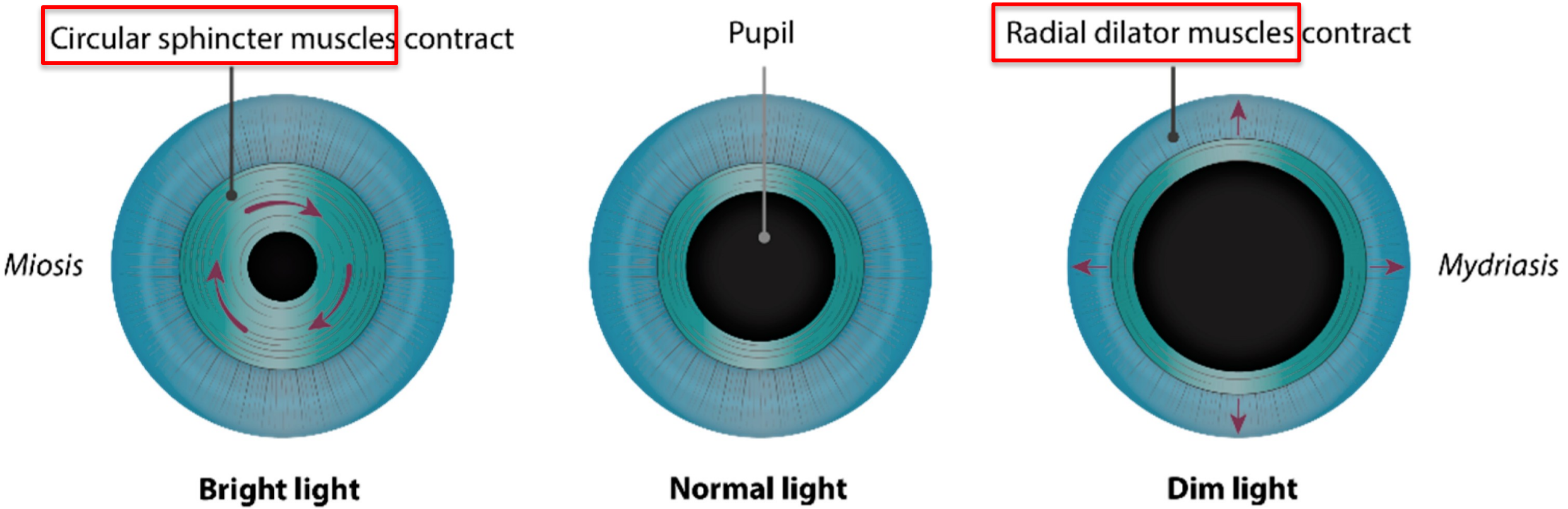
The iris regulates light entry

The iris is like a camera diaphragm!

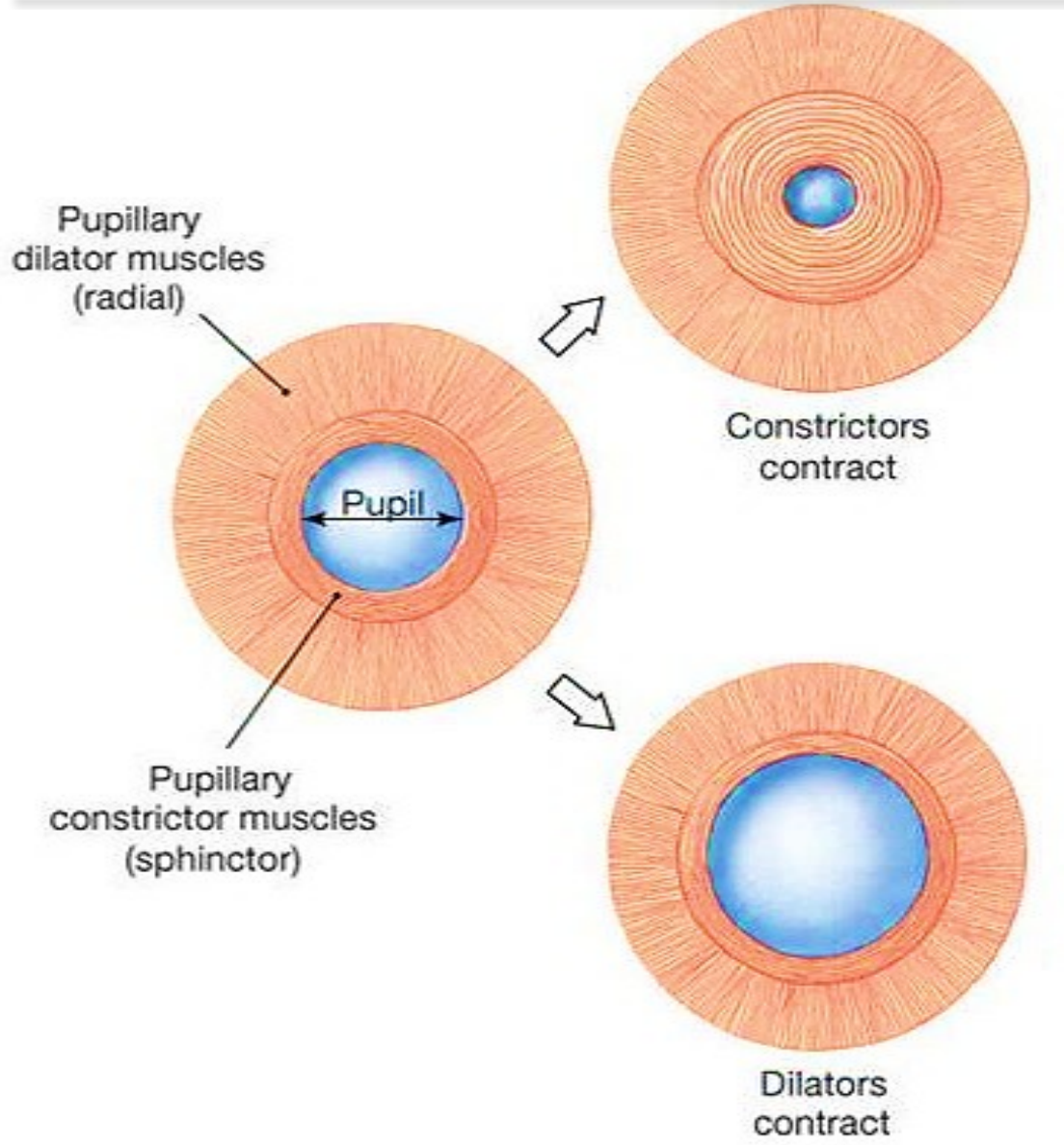
- It adjusts the amount of light reaching the retina.
- Pupil: central opening of the iris.
- Involuntary iris muscles change pupil size; constriction or dilation.



Two muscles control pupil size:



Two muscles control pupil size:



1) Sphincter Pupillae

Circular muscle fibers

- Parasympathetic fibers via the oculomotor nerve.
- Contraction constricts the pupil (miosis).

2) Radial Dilator Pupillae

Radial muscle fibers

- Sympathetic fibers from the superior cervical ganglion.
- Contraction dilates the pupil (mydriasis)

PASSIVE MYDRIASIS

Cholinergic Blocker/ Parasympatholytic:

Atropine

Paralyze the sphincter papillae (constrictor fibers)

→ the dilator fibers unopposed (constriction)

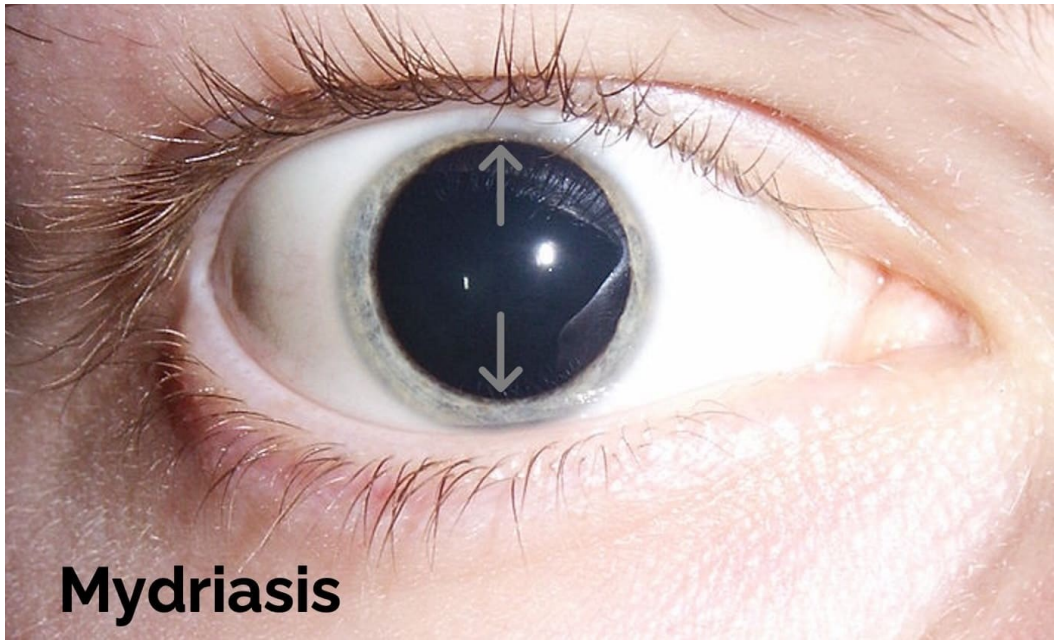
ACTIVE MYDRIASIS

Sympathomimetics/ Mydriatics:

Adrenaline (Epinephrine), Phenylephrine

Stimulate the radial dilator muscle

→ pupil dilation



Atropine

- Alkaloid from Atropa belladonna.
- Blocks acetylcholine at postganglionic parasympathetic nerve endings.



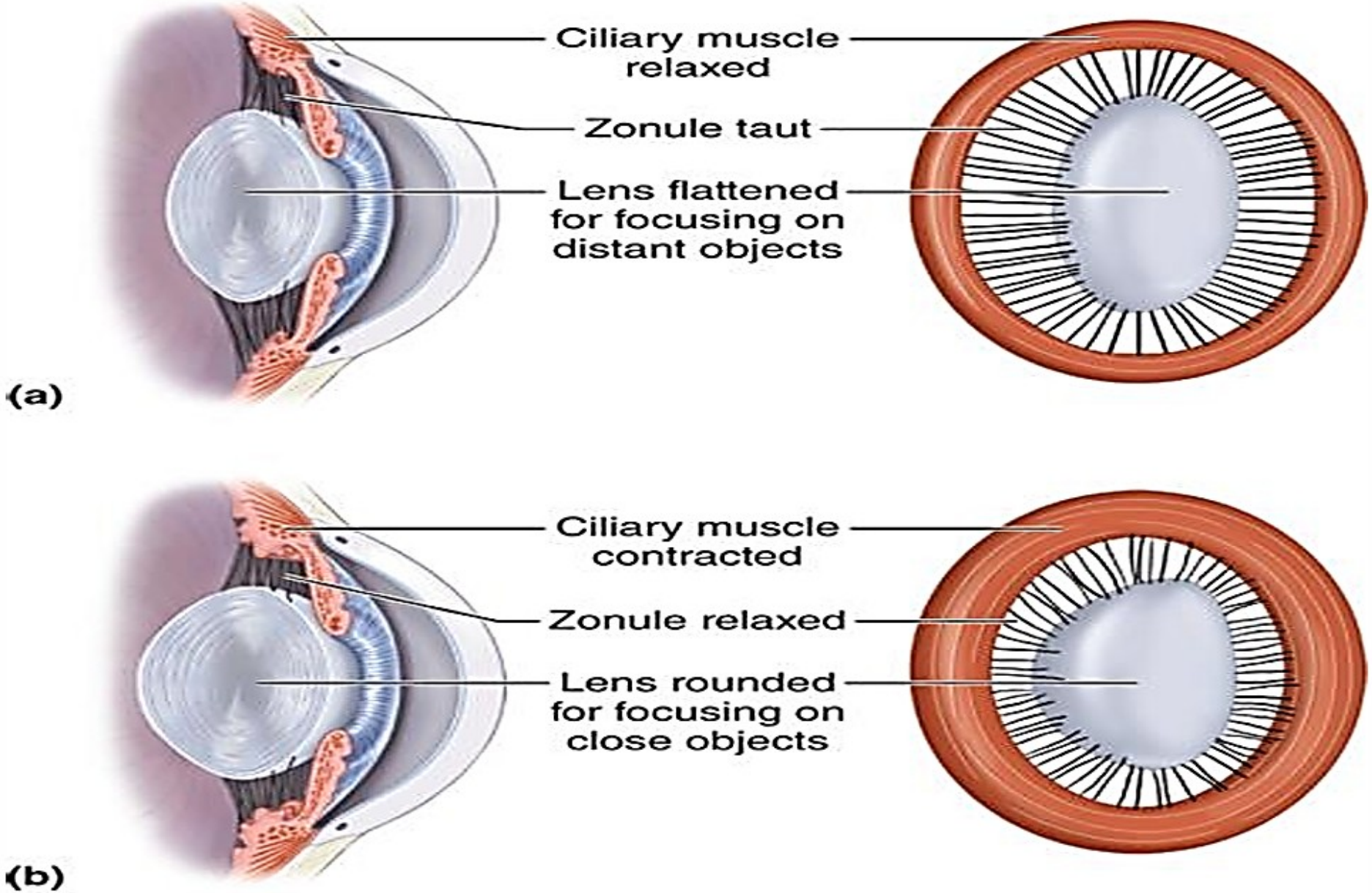


1) CYCLOPLEGIA

Paralysis of the ciliary muscles → inability to focus on nearby objects (accommodate)

2) MYDRIASIS

↑ IOP


Ciliary muscle relaxed
Zonule taut
Lens flattened for focusing on distant objects
(a)
Ciliary muscle contracted
Zonule relaxed
Lens rounded for focusing on close objects
(b)

Atropine

DIAGNOSTIC PURPOSE:

Ophthalmology

- **Eye Exams:** to assess/ measure refractive error without interference from accommodation capacity of the eyes.

- **Surgical Procedures**



Other Anticholinergics for short-acting mydriasis

Uses:

- diagnostic purposes
- pre- or postoperative settings when short-acting mydriasis is needed.

- **Homatropine**

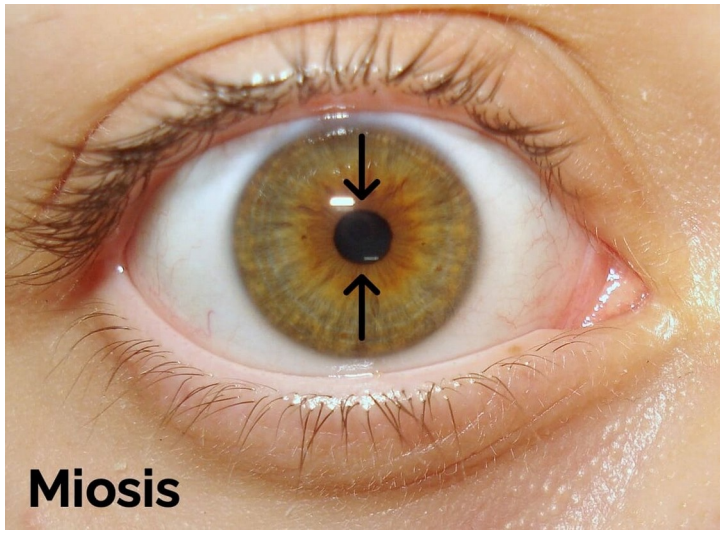


- **Tropicamide** : **SHORTEST HALF-LIFE***





Myotic Agents



DIRECT ACTION

Cholinomimetics

A direct-acting cholinergic receptor agonist → miosis (pupillary constriction).

Pilocarpine

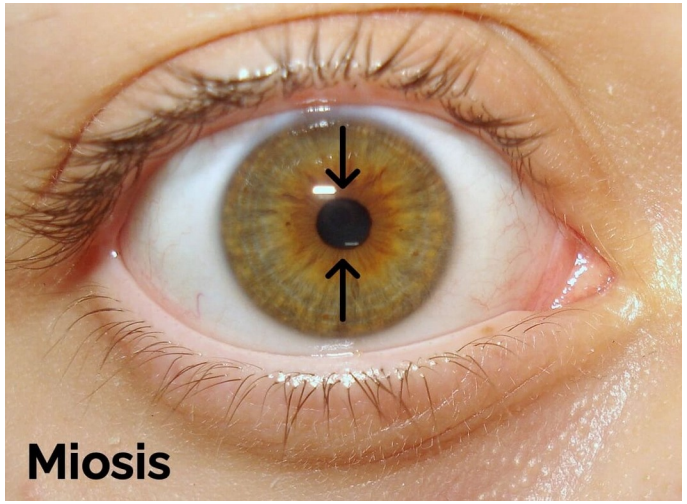
INDIRECT ACTION

Anticholinesterases

- Indirectly through cholinesterase inhibition → miosis
- Can cause spasm of the ciliary muscles
- Used in Glaucoma

Physostigmine and Neostigmine

Drugs producing Passive Miosis



Adrenergic blockading Agents:

- Associated with passive miosis through adrenergic receptor blockade.
- **Phentolamine**

Procedure

BASELINE

For each rabbit: record pupil size, direct and indirect light reflexes, lacrimation, and salivation.

Rabbit A : PILOCARPINE vs. **Rabbit B : TROPICAMIDE**

- Instill two drops into the right eye.
- Wait 10 minutes.
- Compare with the left eye and repeat the baseline observations.

➢ Record your observations and Compare

Observation	Right Eye before Treatment	Right Eye after 10 min	Left Eye
Pupil size			
Direct light reflex			
Indirect light reflex			
Lacrimation			
Salivation			


Thank
You!